

Tejas Dasa

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EDUCATION

The University of Texas at Austin

Expected May 2028

Bachelor of Science, Electrical and Computer Engineering | GPA: 3.92/4.00

Relevant Coursework: Computer Architecture, Introduction to VLSI, Digital Logic Design, Signals and Systems

EXPERIENCE

Electrical Engineering Intern | Infravision Inc.

Mar 2026 – Present

- Profiled a **vision-transformer inference pipeline** and partitioned compute-intensive matrix-multiplication kernels between a companion CPU and FPGA, reducing measured CPU utilization from 75% to 48%.
- Integrated **AXI4-Lite control and DMA** data paths for host requests, input features, weights, and results.
- Analyzed compute, DMA, and Ethernet bottlenecks to guide **CPU-FPGA** partitioning and system optimization.

Undergraduate Research Assistant | Laboratory for Computer Architecture, UT Austin

Jan 2026 – Present

- Architected a cascaded edge-vision inference system combining a fine-tuned **weightless neural network (WNN)** with a confidence-aware gate that escalates difficult inputs to a larger transformer.
- Evaluated WNN designs, fine-tuning methods, and gate models across accuracy, latency, and escalation rate.
- Developing an **FPGA accelerator** for the cascaded pipeline, mapping WNN inference and gating onto pipelined parallel hardware while exploring resource tradeoffs for low power edge deployment.

Hardware Engineer | Texas Aerial Robotics

Sep 2025 – Aug 2026

- Optimized **TensorRT** inference on an NVIDIA Jetson Orin Nano to reduce onboard object-detection latency.
- Designed CAN transceiver PCBs and developed embedded C drivers linking the flight controller with distributed control-surface and telemetry nodes.

PROJECTS

Five-Stage RV32I Processor and Bare-Metal Runtime

[GitHub](#)

- Designed a five-stage RV32I processor in SystemVerilog with forwarding, stalls, traps, and UART I/O.
- Developed a bare-metal C runtime with startup code, linker scripts, memory-mapped UART and timer drivers.
- Verified pipeline and ISA behavior using **UVM**, randomized stimulus, scoreboarding, coverage, and SVA.
- Ran 100+ seeded regressions and validated the core on a **Zynq-7000 SoC** using bare-metal C and UART tests.

Programmable Protocol Emulator ASIC | Jane Street Chip Design Competition

[GitHub](#)

- Architected five independent programmable state machines with deterministic single-cycle execution, implementing **UART, SPI, I²C, JTAG, and CAN** programs within 32-row instruction memories.
- Cross-checked RTL against Python models in cycle-exact lockstep across seven reference programs, 39 randomized programs, and instruction mutations, with zero divergences across all five machines.
- Completed **IHP 130nm implementation in 6x4 Tiny Tapeout** tiles with clean DRC, LVS, and antenna checks.

CUDA Transformer Runtime and KV Cache Manager

[GitHub](#)

- Built a C++/CUDA transformer runtime with custom GPU kernels for prefill and token generation.
- Added block-based KV-cache allocation and memory-aware request scheduling under a fixed memory budget.

LEADERSHIP

Corporate Liaison | IEEE Solid-State Circuits Society, UT Austin

Aug 2026 – Present

- Coordinated industry outreach with semiconductor companies for technical talks and recruitment events.

SKILLS

HDL & Verification: SystemVerilog, Verilog, VHDL, UVM, SVA

Programming & Scripting: C, C++, Python, CUDA, Bash, Tcl, RISC-V Assembly

Tools: Synopsys VCS, Vivado, ModelSim, Verilator, Vitis HLS, Yosys, LibreLane, Magic, KLayout, Linux, Git